

TYRESE MUIGAI

AI-Native Software Engineer | AI Automation Engineer | Cloud Infrastructure Engineer

Nairobi, Kenya | +254 718 004282 | Tyresemuigai09@gmail.com | [LinkedIn](#) | [Kratos Systems](#)

PROFESSIONAL SUMMARY

AI-native software engineer and founder with 2.5+ years of hands-on experience designing and shipping production automation, AI, and full-stack systems for real businesses. Built a 12-workflow lead-to-booking automation pipeline that cut manual sales operations by 80% and brought lead response time under 60 seconds. Currently shaping AI automation strategy and tooling at Wylde International while running Kratos Systems, a software studio delivering full-stack and automation solutions for clients across Kenya. Combines fluency in workflow automation, LLM integration, cloud infrastructure, and full-stack engineering to turn complex business problems into scalable technical systems.

EXPERIENCE

AI Automation Intern | Wylde International | *Jul 2026 -- Present*

- Designing and building n8n automation workflows that integrate Bitrix24 CRM, WhatsApp Business (360dialog), Twilio, and Google Calendar to reactivate and qualify a 21,000-contact lead database.
- Authored the technical proposal and architecture for a QR-code-driven lead capture and discovery-call booking system, adopted by the marketing and strategy teams.
- Involved in technical planning for a full brand rebuild and new company website, owning architecture and front-end implementation from brief through production.
- Scoping and building a custom internal strategy tool to help client SMEs move from strategy to execution, closing the implementation gap that leaves most business strategies unacted on.

- Advising leadership on where automation, AI, and integration investment will move client-acquisition and internal-operations metrics.
-

Software & Automations Engineer (Contract) | Social Funnel | *Jan 2026 -- Present*

- Built AcquisitionOS, a bespoke lead-to-booking automation engine, using n8n and the OpenAI API to run real-time lead qualification, CRM logging, and calendar scheduling end to end.
 - Implemented a 60-second automated response protocol that reduced manual sales operations by 80% and improved lead conversion through a nurturing pipeline utilising context-aware AI messaging.
 - Ongoing contract to customise and extend the AcquisitionOS platform for new clients that Social Funnel onboards, adapting workflows, CRM integrations, and qualification logic per engagement.
-

Founder and Lead Engineer | Kratos Systems | *Apr 2025 -- Present*

- Founded Kratos Systems as a one-person software studio to ship serious, production-grade engineering for Kenyan and African businesses.
 - Designed, built, and deployed Ashgate, a full-stack real estate ecosystem platform, as sole engineer from data architecture through cloud production deployment.
 - Delivering full-stack applications end to end including React/TypeScript frontends, REST APIs, role-based admin dashboards, and POS/inventory systems for client businesses.
-

Full-Stack Software Engineer (Internship) | SokoFresh EA | *Apr 2024 -- Dec 2024*

- Led development of a comprehensive cold storage management system handling real-time inventory tracking for tools, spare parts, and battery management across 15+ cold storage units and NTUs in Kenya.
- Built a robust RESTful API architecture using Laravel 12 and Eloquent ORM, managing complex relationships across inventory, batteries, cold storage units, and user management.

- Developed responsive React.js frontends with dynamic Leaflet.js map integration for real-time GPS location tracking of cold storage units across the country.
 - Designed and optimised a normalised PostgreSQL schema with full migration versioning, foreign key constraints, and query performance tuning for production scale.
 - Implemented end-to-end user authentication and authorisation using Laravel Sanctum across all system roles.
-

Data Engineer and UI/UX Contributor | GlycoSafe (JHub Africa) | 2024

- Owned data acquisition, cleaning, and preprocessing for the platform's computer vision pipeline (YOLOv11 + ViT-B/16), ensuring high-quality labeled datasets for food segmentation and classification.
 - Designed and shipped Rafiki, the platform's RAG-grounded conversational AI assistant, integrating a FAISS knowledge base, QLoRA-adapted Llama 3, and a deterministic clinical safety-override layer.
 - Contributed UI/UX design decisions that shaped user interaction across the mobile application.
-

LEADERSHIP AND COMMUNITY

- **Mastercard Foundation Associate** | Mastercard Foundation via Shortlist | *May 2026 -- Dec 2027 (active)*
Selected for the Mastercard Foundation Associates Programme to gain hands-on industry experience with leading organisations across Africa and globally. Supports emerging professionals through work-integrated learning, remote collaboration, mentorship, and practical experience in real-world business and technology environments. Placements ongoing through December 2027.
- **Chairperson, Entrepreneurship Club** | Mangu High School | Led and organised entrepreneurship activities across school events, coordinating student-run food businesses during school functions and hosting events that built practical commercial and leadership skills among members.
- **Pioneer Public Relations Officer** | Mangu High School Interact Club | Led outreach programs and public relations efforts for Rotary-affiliated community and charity initiatives.

- **Active Member** | Rotary Club of Lavington | Leading community service initiatives and cross-functional local development projects; managing professional networking within the Rotary International network.
 - **Volunteer** | Malindi Red Cross | Supported disaster response and community health programs, including emergency-preparedness awareness efforts.
-

EDUCATION

BSc. Computer Science | Jomo Kenyatta University of Agriculture and Technology (JKUAT) | Second Class Upper Honours

Mangu High School | Kenya Certificate of Secondary Education | Grade A-

CERTIFICATIONS AND CONTINUING EDUCATION

- Google Cloud Certified: Associate Cloud Engineer
 - AWS Certified Solutions Architect (Associate, in progress)
 - Anthropic Academy: AI Fluency, Framework and Foundations
 - Claude Code 101, Introduction to Model Context Protocol, Model Context Protocol: Advanced Topics
 - Claude with Amazon Bedrock & Claude on Google Cloud (in progress)
-

TECHNICAL SKILLS

Programming: Python, Swift, Go, Java, JavaScript/TypeScript, PHP, Bash, Rust

Cloud and Infrastructure: Google Cloud Platform (Proficient), AWS, Docker, Kubernetes, Terraform, Ansible

Automation and AI Integration: n8n, LangChain, OpenAI API, Claude API, HuggingFace, Make.com, Zapier, ManyChat

Software Development: Full-stack web (React, Next.js, Laravel, Django), REST API design, microservices, system architecture

Databases: PostgreSQL, MySQL, MongoDB, Firebase, Redis, SQLite

AI and ML: PyTorch, QLoRA/PEFT fine-tuning, RAG systems, YOLOv8/v11, NumPy, SciPy, Pandas, Matplotlib

LANGUAGES

English (Fluent) | Swahili (Fluent) | French (Intermediate)

REFERENCES

Available upon request.